

Software engineering assignment

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1 Introduction- My research area

The topic of my research is designing software testing framework for ethical alignment of AI systems. In this research, we aim to study the existing challenges related to AI alignment in the industry and propose a framework later on how these challenges can be tested and mitigated. As a part of my research thesis, I have started with a study of "AI Alignment for Ethical Compliance and Risk Mitigation in Industrial Applications". This was an interview study where the interview participants were presented with some frameworks related to ethical AI and risk assessment for industrial applications.

The current study that I'm performing is an interview study to understand opportunities and challenges for continuous safety assurance in Autonomous Driving Systems- through the lens of contract-based assurance and operational testing. Although we do not focus directly on ethical aspects of AI alignment, but this study sets to bring in some interesting insights related to safety alignment.

In order to design or propose a testing framework, I have explored existing ethical alignment frameworks, for example, Guidelines for trustworthy AI by the European Commission [3], ECCOLA framework [8], and a framework on double-edge components [5].

2 Lecture principles

I have chosen quality assurance and testing as two concepts/topics that I would like to relate to my research and discuss.

Starting with quality assurance, I think this aligns very well with my research. Since my idea is to design or build a testing framework for making sure that the system is ethically aligned, the goal calls out for considering quality assurance. so it is really important for me to consider how quality assurance is performed, and what factors need to be catered. after attending the course lectures, I got interesting insights into both verification and validation and how to relate it to requirements engineering. I think the validation process of my research would include the studies that I'm performing currently with the in-

dustries, understanding about the problem that would help to build the right testing framework.

Based on the discussion in the lecture about methods of verification and validation, from static, dynamic and to process evaluation techniques, I think the design of the testing framework for my thesis would lie around static techniques, atleast to start with.

Given the complexities of the AI systems, use of static techniques will play an important role in making sure whether proper guidelines have been followed or not, while deploying the AI systems, and if they align well with the intentions of the stakeholders.

Process evaluation techniques would also be valuable for assuring how the software has been produced, and whether ethical guidelines for AI system(may be from the EU AI Act) have been followed or not while designing and building the system.

Another concept that I would like to relate to my research is Behavioral Software Engineering(BSE). I think this is a very important aspect which I should during my research thesis, because understanding the ethical values for AI systems is somehow related to the psychology of the people defining them. So the meaning of ethics can of course vary.

The ways three units of analysis in BSE and presented, resonates very well with the levels of AI alignment described in a similar fashion(Individual/ organizational/ national/ global) [4].

When considering cognitive bias, I agree that confirmation bias is an important aspect to be considered for my studies, as people tend to follow ethics based on their preconceptions, and what is considered for one stakeholder, might not be relevant at all for another one. So while testing AI systems for ethical alignment, it is important to have the correct reference of what is perceived as "ethical" among people and under given circumstances/environment.

3 Guest Lecture principles

Starting with Frattini's lecture contents, thinking about requirements engineering is very important from the beginning. Relating the techniques of requirements engineering to my research topic, I think all of them need to be thought carefully. Starting with stakeholder elicitation, it is really important to elicit stakeholders, since the alignment thought can vary between different stakeholders, even if they are within the same organization.

Goal modelling is another aspect that I find very relevant for my project. It can be mapped to how well the intents of AI alignment have been defined and in what aspects. For example, consider an AI system for which the intent is to assist a robot in identifying packages on a production line, but at the same time, it is incorporated with ethical values of ensuring a worker's safety and working in collaboration with it, and not harming them.

From Per Langberg's slides discussions, I found the discussion on V-model software development quite interesting. Thinking about designing a testing

framework to evaluate an AI system, and then having components of AI within different phases of SDLC for that framework itself, places me in a very difficult situation to ponder about. It feels like working on a loop that might result in a deadlock of evaluation guidelines. It needs to be designed curatively keeping several aspects in mind.

If this is the case, the credibility of the testing framework needs to be explained on different aspects to ensure the trustworthiness of the evaluation results.

4 Data scientists, software engineers, and AI engineers

I have studied these chapters [7] and [6], and will answer the following questions based on that: Differences between data scientists and ML engineers as described in the book:

Data scientists typically have strong backgrounds in statistics and machine learning, often with advanced degrees. Their work focuses on developing and improving models through tasks such as feature engineering, model design, and hyperparameter tuning, as well as collecting and cleaning data. They often use an exploratory, research-oriented workflow with tools like Jupyter notebooks and evaluate models mainly based on accuracy, with less emphasis on factors such as latency, scalability, or cost.

Software engineers, by contrast, focus on building software products that meet user needs within time and budget constraints. Their work includes gathering requirements, designing systems, implementing, testing, deploying, and maintaining software. They must balance trade-offs between usability, scalability, maintainability, security, development time, and cost. Software engineering education therefore covers areas such as requirements engineering, software design, testing, distributed systems, and security, often culminating in a practical project for a customer.

I agree with the distinction as mentioned in the book because both of these profiles carry different roles and responsibilities. Production ML indeed requires a system-wide view, not only a model-centric view. Relating to my research area, working on a specific AI system might be the role of a data scientist, but considering the ethical aspects of AI alignment, understanding the intentions of the stakeholders would be the responsibility of a software engineer.

I don't think that data scientists and software engineers will remain completely distinct roles; of course, to some extent, yes, and both will definitely need to learn more of each other's skills. Both roles need to go hand in hand, learn each other's work on a broader level to have broader knowledge (T-shaped), as discussed in the chapters. Both roles are incomplete without each other at a certain stage. A data scientist does need to have an understanding of the software engineering discipline when designing the ML model, and vice versa, a software engineer should also have some understanding of the model, data, and

features on a broader level for better decision-making. But at the same time, some distinction of expertise in their specific roles will definitely be there.

The term AI engineering can be described as partly distinct but primarily an extension of software engineering for AI-enabled systems. It includes MLOps and model engineering, such as deploying, monitoring, and automating ML pipelines, but it should go beyond these areas.

In the book chapter, it is mentioned that MLOps can remain too model-centric if it focuses only on training, deployment, and monitoring. AI Engineering should instead consider the entire system: how AI components interact with traditional software, users, data, and the surrounding environment.

I think that the foundation models, AI coding assistants, RAG systems, prompt pipelines, and agentic workflows will make the boundaries thinner between these roles. Software engineers may think of reusing an existing model with the help of agents or prompts, instead of building it from scratch, let's say.

5 Paper analysis

5.1 1st paper- Ethical Challenges and Frameworks in AI-Driven Software Development and Testing [2]

5.1.1 Core ideas and their SE importance:

The core ideas of the paper are regarding the ethical aspects and ethical frameworks to be considered during the design, development and testing of the AI systems. The paper talks about why ethical considerations are important for the development of the AI-systems. It also shed some light on some of the existing ethical frameworks, and discussed about the common core principles among them, such as human-centered design, accountability, transparency, fairness, non-discrimination, privacy and security. It also presented strategies which can be utilized for embedding ethical considerations into AI-applications right from the design phase.

5.1.2 Relation to my research

The core idea of the paper is highly relevant for my research topic. One of the most interesting aspect I found in this paper is their discussion about the tools and methodologies that support ethical AI development. As a software testing framework designer for ethical AI alignment, it is important for me to know about these practical application in AI system development. They have also emphasized on relevance of this topic to software development and testing, for example, discussing about technical strategies on how we can detect and mitigate bias.

5.1.3 Integration into a larger AI-intensive project

Let us consider an example of autonomous driving system. A company has recently acquired an autonomous driving cars firm wants to emphasize the responsible AI angle in it. So the concepts from this paper goes very well into the vision of this company, where the paper demonstrates the important aspects like ethical impact assessments while designing models, inclusive design teams involving ethicists, sociologists and domain experts, which the company should make sure to consider. Also, focusing on balancing the dataset can be really important, as the car's backend models can rely quite heavily on the trained dataset, so we need to make sure all the edge cases are covered. From my research, the testing framework when ready to test the AI alignment, can be used to test several aspects of this project on AI alignment, testing whether the models are aligned with the stakeholders intentions or not, for example braking in good time, when for example, a kid sitting on mini scooter and crossing the road before a pedestrian crossing. Also, if a dataset has no reports of accident prone zone, is the car capable of adapting to the speed based on the road conditions, irrespective of no accidents reported yet, but highly likely given the curvy roads and residential area ahead.

5.1.4 Adaptation of my research

I would take inspiration from this paper on how to incorporate responsible AI in software testing. The paper discusses about the responsible AI development life cycle, as they have quoted in the paper: "Testing AI with AI to make a system as Responsible AI". Using AI driven test automation models for requirement analysis, test planning, test case development(getting the guardrails implemented for testing ethical AI alignment), test execution and test cycle closure. So I would like to adapt my research by using automated AI-driven testing framework for testing ethical AI alignment and validate the results.

5.2 2nd paper- An AI-driven Requirements Engineering Framework Tailored for Evaluating AI-Based Software [1]

5.2.1 Core ideas and their SE importance:

The core idea of the paper is that Requirements Engineering(RE) for AI-based software(AIS) is different from that for traditional systems. This paper proposes a framework that works on domain knowledge from RE and applies explainable AI for AIS for pedestrian and aircraft detection use cases, which utilizes vision-based perception. The authors have evaluated the proposed framework via four distinct metrics such as conceptual alignment, specification alignment, human-likeness, and performance generalization. This paper highlights the importance of Requirement Engineering for AI(RE4AI).

5.2.2 Relation to my research

This paper relates quite well to my research project. The paper talks about conceptual and RE specification alignment in the evaluation of their framework(W-AIS), while my research also focuses on alignment, but from an ethical perspective. It is interesting to study how they have incorporated explainable AI to align AI systems' perceptions of the RE specifications. I can utilize this concept of using XAI while designing my framework for testing ethical alignment. In terms of methodology, I found it interesting how they have presented and explained their methodology very well, by describing their workflow diagrammatically, and explaining each step with the help of an algorithm.

5.2.3 Integration into a larger AI-intensive project

Let us consider an example of an AI system designed for tumor detection in the medical domain. In this case, the core idea is the identification of the tumor/tumors based on the image of the specific parts of the body. The framework proposed in the paper promises better classification of pedestrians and aircraft based on RE specifications for AI systems, given the image dataset.

When thinking about how my research could relate to this bigger project, I think ethical alignment is very important in the medical domain. Factors like trustworthiness, explainability, and accountability are really important to consider in such life-critical scenarios. A testing framework for ethical alignment would evaluate this AI- based medical solution to check for these factors, among others.

5.2.4 Adaptation of my research

I can adapt my research by getting some inspiration from the paper's idea of building a framework by utilizing RE specifications for AIS. To build a testing framework, it is important to think about RE specifications, and this paper has given an idea to think around that concept. Using explainable AI to justify the logic behind AI system perception/action, would help to build a robust testing framework, which can along with each passing or failing test case, describe the reasoning behind it.

6 Evidence, originality, and GenAI-use transparency

The main papers and the sources are listed in the reference list.

The different types of resources are:

- Article- [5], [8], [4]
- Book chapter [7], [6]
- conference proceedings - [2], [1]

- Report - [3]

I trust all the sources mentioned in the reference list, as they are published papers, articles, and reports at trustworthy places.

ChatGPT was used only for analyzing some section of paper 2 [1] for validating my summary, but I found some analysis was missing in the findings shared by chatGPT, so I have presented my own summarization here as well. Apart from it, no use of GenAI tools has been made anywhere else.

References

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